INNOVIBE
Innovation. Energy. Impact.

SMART INDIA HACKATHON 2026

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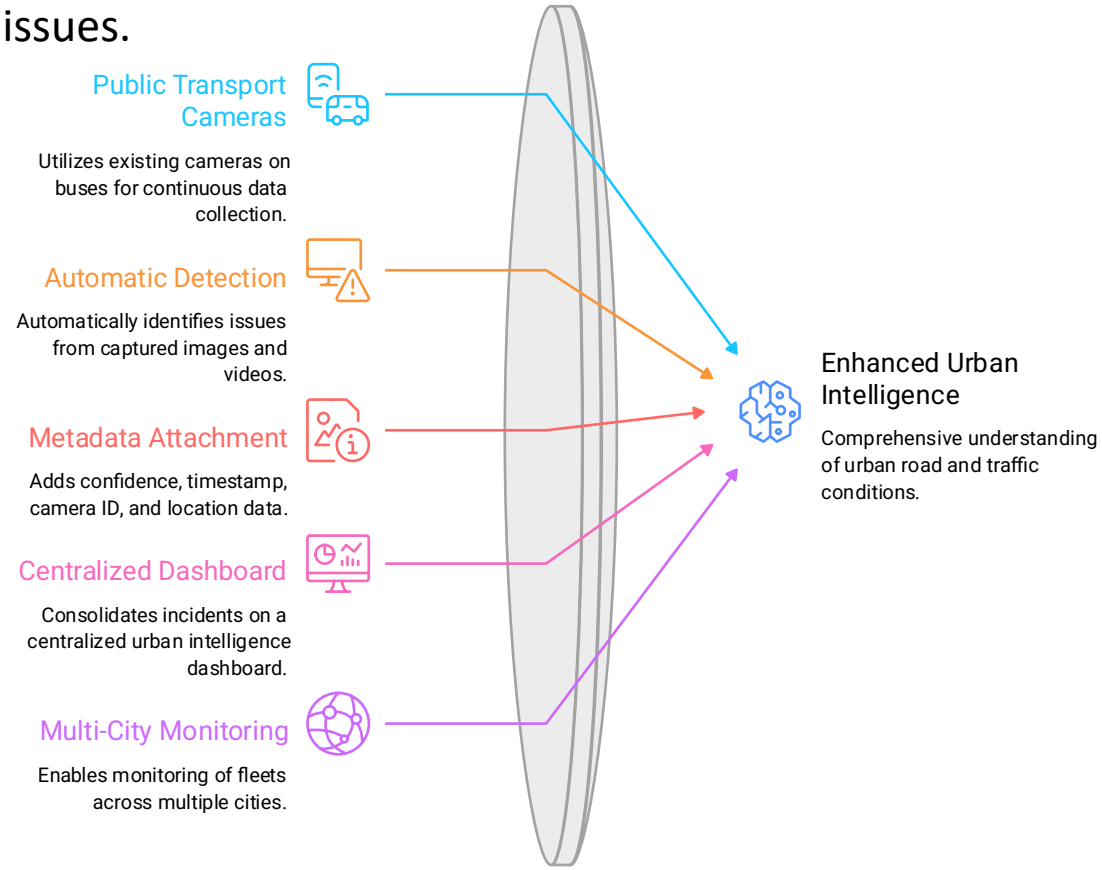
- **Problem Statement ID - 26124**
- **Problem Statement Title –**

  AI-Powered Mobile Urban Intelligence Platform
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - 183556**
- **Team Name – InnoVibe**

Fleet Vision

- Fleet Vision converts public transport buses into mobile urban sensing units by using their onboard cameras to identify urban road and safety issues.
- **4 AI Detection Modules :**
  - **Road Defects —** Detects potholes, waterlogging, cracks, and damaged road surfaces from visual data.
  - **Traffic Density —** Detects and counts vehicles to estimate traffic volume and congestion levels.
  - **Missing Infrastructure —** Identifies missing/damaged zebra crossings, traffic signs, and lane markings.
  - **Hit & Run Detection —** Detects road incidents, tracks the involved vehicle, and extracts its number plate when legible.
- The repository models each bus with Front, Left, Right and Rear cameras and supports multiple cities with a fleet dashboard.

FleeVision's Urban Sensing Network

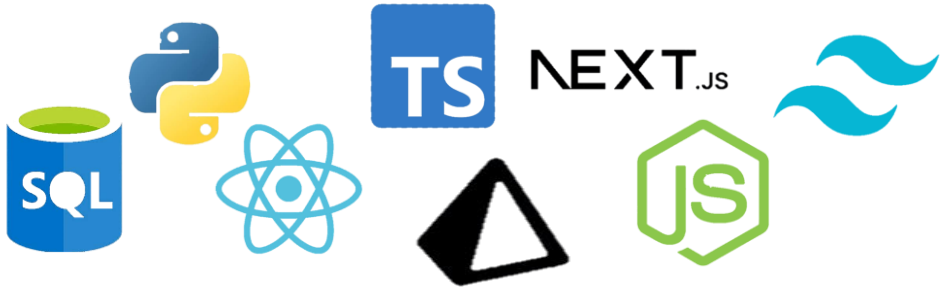


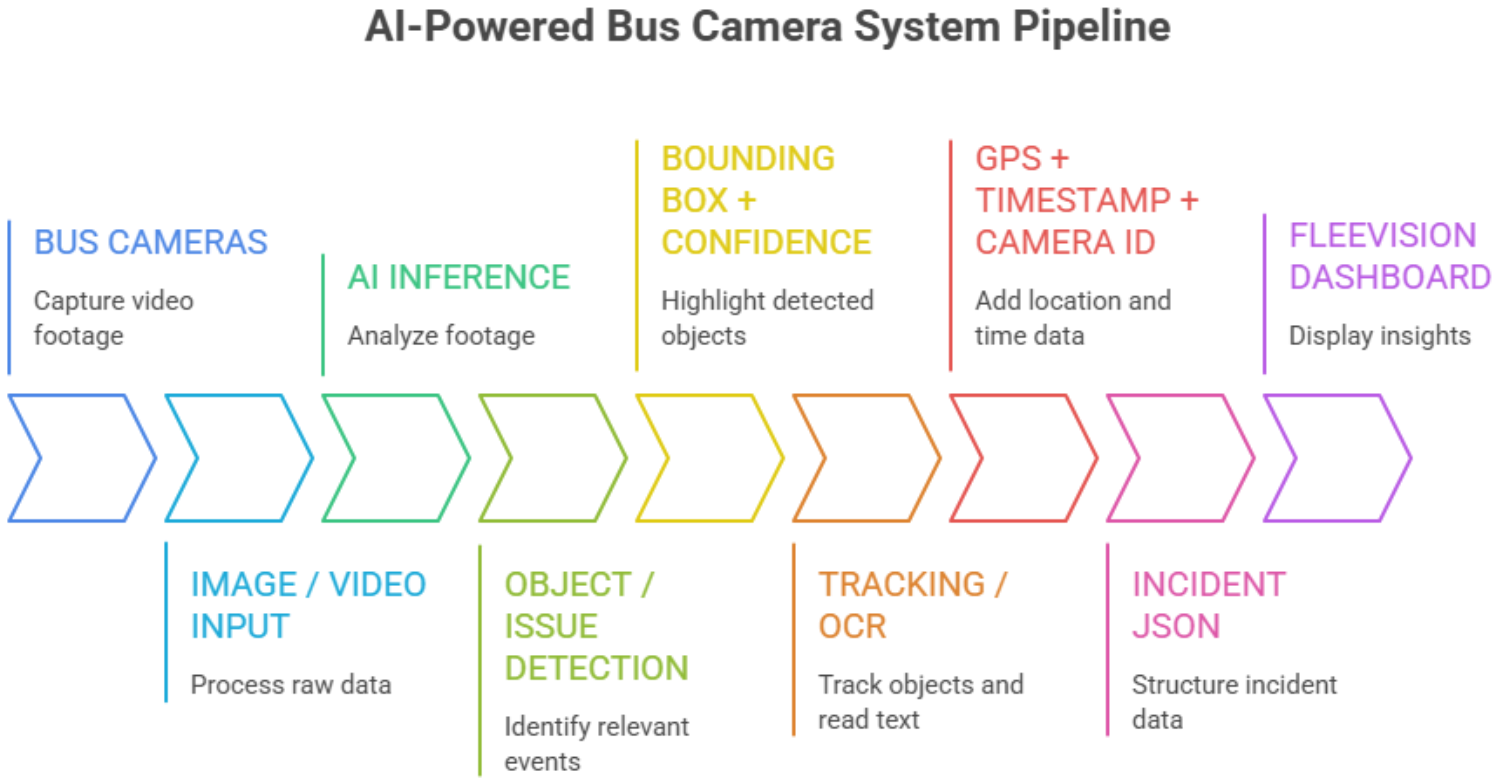
Prototype: www.Fleetvision.com

TECHNICAL APPROACH

❖ Tech Stack :

- ➢ **Frontend :**
  React, Next.js, HTML/JSX, Tailwind CSS, Zustand, Leaflet/OpenStreetMap, Recharts
- ➢ **Backend :** TypeScript, Node.js
- ➢ **AI / ML :** GLM-5.2, YOLOv11, Byte Track, OCR
- ➢ **Data :** SQLite, JSON
- ➢ **APIs :** Next.js API Routes
- ➢ **Offline Support :**
  local image video processing





FEASIBILITY AND VIABILITY

FEASIBILITY :

- Proven AI Stack — YOLO v11, Byte Track, OCR & GLM Vision are mature, production-tested technologies.
- Existing Infrastructure — Reuses cameras already installed on public buses, reducing hardware requirements.
- Modern Software Stack — Next.js, React, TypeScript & OpenStreetMap enable practical, maintainable development.
- Clear Demo → Live Path — Current demo can transition to production with a backend, database & reliable data pipeline.
- Scalable Deployment — Same core system can support the proposed 8-city, 12-bus-per-city pilot model.
- Manageable Compliance — Privacy, data retention and access control are required but are solvable implementation challenges.

VIABILITY :

- Cost-Effective Model — Existing bus infrastructure avoids the high cost of installing roadside CCTV across cities.
- Multi-Department Value — Supports road maintenance, traffic management & police through one platform.
- Easy to Scale — New cities and detection categories can be added without rebuilding the entire system.
- Long-Term Sustainability — Ongoing costs are mainly software, cloud/edge computing and server infrastructure.



IMPACT AND BENEFITS

Impact :

Road Safety

- Early detection of potholes and road damage before they cause accidents.
- Automatic identification of missing or damaged traffic signs.
- Hit-and-run evidence with image, GPS location, timestamp and camera ID.
- Faster investigation and improved road-safety response.

Governance & Decision-Making

- Centralized GIS map for road defects, traffic and incidents.
- Repair priorities based on confidence, location and detection frequency.

Coverage & Cost Efficiency

- Existing bus cameras can be used for continuous road monitoring.
- Buses provide wider road coverage than fixed CCTV systems.

Benefits :

Citizens :

Safer roads, improved road conditions and better accountability .

Road & Municipal Authorities :

Continuous road monitoring, prioritized repair lists and reduced manual inspection.

Traffic Police :

Evidence images with GPS, timestamps and camera metadata to support hit-and-run investigations.

Government :

A scalable smart-city platform supporting data-driven road management and the SIH problem statement.

RESEARCH AND REFERENCES

OBJECT DETECTION RESEARCH

YOLOv11 :

Research on real-time object detection for identifying vehicles, potholes and road defects from camera footage.

Link:
https://docs.ultralytics.com/models/yolo11

VEHICLE TRACKING RESEARCH

Byte Track :

Multi-object tracking algorithm for tracking vehicles across video frames, including partially occluded objects.

Link :
https://arxiv.org/abs/2110.06864

OCR & VISION RESEARCH

Paddle OCR & GLM Vision :

Research on number-plate text recognition and AI-based visual understanding of road scenes.

Link:
https://github.com/PaddlePaddle/PaddleOCR

ROAD DAMAGE DATASET

RDD2022 :

Annotated road images containing cracks and potholes for training and evaluating road-damage detection models.

Link :
https://github.com/sekilab/RoadDamageDetector

TECHNICAL STANDARDS

OpenStreetMap & Geo JSON :

Open mapping and geographic data standards for visualizing road defects, GPS locations and incidents on a centralized map.

Link :
https://www.openstreetmap.org/